# Chapter 3: Expectation of Random Variables

Probability and Statistics for Science and Engineering with Examples in R

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# (3) Covariance and Independence

Covariance of $X_1$ and $X_2$:

$$\mathrm{Cov}\,(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)]$$

If $X_1$ & $X_2$ are independent, then

$$\mathrm{Cov}\,(X_1, X_2) =$$

Alternative form of the covariance:

$$\mathrm{Cov}(X_1, X_2) = E(X_1 X_2) - \mu_1 \mu_2$$

# Theorem 3.2

$X$ and $Y$ are independent $\underset{\not\Leftarrow}{\Rightarrow}$ $\mathrm{Cov}(X, Y) = 0$

<u>Proof</u>

If $X$ and $Y$ are normal r.v.'s, then the iff condition is satisfied.

<u>Theorem 3.3</u>

(1) $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$

<u>Proof</u>

(2) $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$

<u>Proof</u>

## Theorem 3.3 (continued)

(3) $\mathrm{Cov}(aX + bY, cX + dY)$

$$= ac\mathrm{Var}(X) + bd\mathrm{Var}(Y) + (ad + bc)\mathrm{Cov}(X, Y)$$

## Proof

# Theorem 3.3 (continued)

(4) If $X_i$ is a r.v. and $a_i$ is constant, $i = 1, \cdots, n$, then

$$\mathrm{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \mathrm{Var}(X_i) + 2\sum_{i<j}\sum a_i a_j \mathrm{Cov}\left(X_i, X_j\right)$$

# Correlation Coefficient

Correlation coefficient:

$$\rho = \frac{\mathrm{Cov}(X_1, X_2)}{\sigma_{X_1}\sigma_{X_2}}$$

$$-1 \leq \rho \leq 1$$

<u>Proof</u>

<u>Example 5.2:</u> Find the marginal distribution of $X$ and $Y$, and $P\left(\frac{1}{2} \leq Y \leq \frac{3}{4}\right)$.

$$f(x,y) = \begin{cases} \frac{3}{8}(x+y^2), & 0 \leq x \leq 2, 0 \leq y \leq 1. \\ 0, & \text{otherwise} \end{cases}$$

$f_1(x) =$

$f_2(y) =$

<u>Example 3.14</u> (continued): Find the correlation coefficient of $X$ and $Y$.

$$f_1(x) = \begin{cases} \frac{1}{8}(3x+1), & 0 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

$E(X) =$

$E(X^2) =$

$Var(X) =$

# Example 3.14 (continued)

$$f_2(y) = \begin{cases} \frac{3}{4}(y^2 + 1), & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$E(Y) =$

$E(Y^2) =$

$Var(Y) =$

# Example 3.14 (continued)

$$f(x,y) = \begin{cases} \frac{3}{8}(x+y^2), & 0 \le x \le 2, 0 \le y \le 1. \\ 0, & \text{otherwise} \end{cases}$$

$\mathrm{E}(XY) =$

$Cov(X,Y) =$

$\rho =$

- Let $X_1, \cdots, X_n$ be $n$ r.v.s & $a_1, \cdots, a_n$ be constants. Then

$$Y = a_1 X_1 + \cdots + a_n X_n = \sum_{i=1}^{n} a_i X_i$$

is called a <u>linear combination</u> of the $X_i$'s.

eg) If $a_1 = \cdots = a_n = 1$, then $Y = \sum_{i=1}^{n} X_i$

If $a_1 = \cdots = a_n = \frac{1}{n}$, then $Y = \bar{X}$

- Let $X_i$ have mean $\mu_i$ & variance $\sigma_i^2$ for $i = 1, 2, \cdots, n$. Then
  - $E(a_1 X_1 + \cdots + a_n X_n) = a_1 E(X_1) + \cdots + a_n E(X_n)$
    i.e., $\mu_Y = \sum_{i=1}^{n} a_i \mu_i$
  - If $X_1, \cdots, X_n$ are independent,
    $\mathrm{Var}(a_1 X_1 + \cdots + a_n X_n) = a_1^2 Var(X_1) + \cdots + a_n^2 Var(X_n)$
    i.e., $\sigma_Y^2 = \sum_{i=1}^{n} a_i^2 \sigma_i^2$

## Example 3.15

Prices of 3 different brands of toothpastes: \$2.40, \$2.70, \$2.90 per pack

$X_i$: total amount of brand $i$ purchased on a particular day, $i = 1, 2, 3$

$X_i's$ are independent r.v.'s with $\mu_1 = 500, \mu_2 = 250, \mu_3 = 200$;

$$\sigma_1 = 50, \quad \sigma_2 = 40, \quad \sigma_3 = 25$$

The revenue from sales: $Y = 2.4X_1 + 2.7X_2 + 2.9X_3$

What are the mean and standard deviation of $Y$?

$E(Y) =$

$Var(Y) =$

$\sigma_Y =$

Special case: $n = 2$ , $a_1 = 1$ & $a_2 = -1$ . Then

$$\mathrm{Y} = a_1 X_1 + a_2 X_2 = X_1 - X_2$$

$$E(X_1 - X_2) = \mu_1 - \mu_2$$

If $X_1$ & $X_2$ are independent, then $\mathrm{Var}(X_1 - X_2) = \sigma_1^2 + \sigma_2^2$.

<u>Example 3.16</u> The tar contents of the cigarettes from two different brands are known to be different. Let $X_1$ and $X_2$ be the tar contents of the cigarettes from Brand 1 and Brand 2, respectively. $X_1$ and $X_2$ are independent.

Let $\mu_1 = 15$ mg, $\mu_2 = 17$ mg, $\sigma_1 = 1.1$ mg and $\sigma_2 = 1.4$ mg.

$\Rightarrow E(X_1 - X_2) =$

$Var\ (X_1 - X_2) =$

$\sigma_{X_1 - X_2} =$

If $X_1, \cdots, X_n$ are independent and $X_i \sim N(\mu_i, \sigma_i^2)$ for $i = 1,2, \cdots, n$,

$$Y = a_1 X_1 + \cdots + a_n X_n \sim N(a_1\mu_1 + \cdots + a_n\mu_n, a_1^2\sigma_1^2 + \cdots + a_n^2\sigma_n^2)$$

$$X_1 - X_2 \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$$

<u>Example 3.17</u>

From <u>Example 3.15</u>, suppose $X_1, X_2$ & $X_3$ are normal. Then

$$Y = 2.4X_1 + 2.7X_2 + 2.9X_3$$

$$Y \sim N(2.4\mu_1 + 2.7\mu_2 + 2.9\mu_3, 2.4^2\sigma_1^2 + 2.7^2\sigma_2^2 + 2.9^2\sigma_3^2)$$

# Example 3.18

Let $X_1, \cdots, X_4$ be independent normal random variables and $X_i$ distributed as $N\left(\mu_i, \sigma_i^2\right)$ for $i = 1, \cdots, 4$.

(a) Find $P(\bar{X} < 11)$ when $\mu_1 = \cdots = \mu_4 = 10$ and $\sigma_1^2 = \cdots = \sigma_4^2 = 2$.

# Example 3.18 (continued)

Let $X_1, \cdots, X_4$ be independent normal random variables and $X_i$ distributed as $N\left(\mu_i, \sigma_i^2\right)$ for $i = 1, \cdots, 4$.

(b) Find $P(4X_1 + X_2 - X_3 - 4X_4 < 3)$ when $\mu_1 = \cdots = \mu_4 = 10$ and $\sigma_1^2 = \cdots = \sigma_4^2 = 2$.

# Example 3.18 (continued)

Let $X_1, \cdots, X_4$ be independent normal random variables and $X_i$ distributed as $N\left(\mu_i, \sigma_i^2\right)$ for $i = 1, \cdots, 4$.

(c) Find $P(35 < X_1 + \cdots + X_4 < 45)$ when $\mu_1 = \mu_2 = 8$, $\mu_3 = \mu_4 = 12$, $\sigma_1^2 = \sigma_2^2 = 1$ and $\sigma_3^2 = \sigma_4^2 = 2$.

# Example 3.19

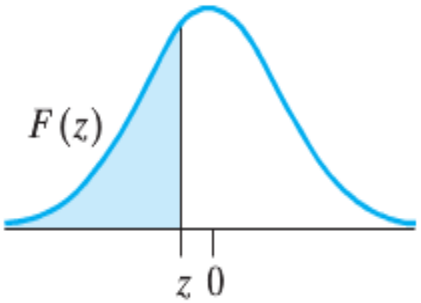
Let $U$ and $V$ be independent N(0,1) r.v.'s.

Set $Z = \rho U + \sqrt{1-\rho^2}V, -1 < \rho < 1.$

(a)Determine the distribution of $Z$.

(a) Find the correlation of $U$ and $Z$.

# Table A.1 Standard Normal Distribution Table



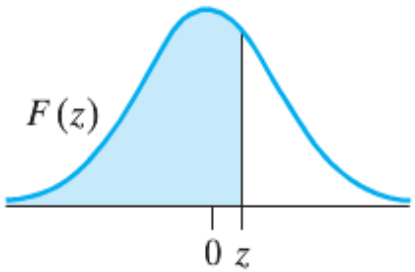

(Area under $z$ curve to the left of $z$)
$= P(Z < z) = P(Z \leq z)$
$= \Phi\ (z)$

| $z$ | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| −3.4 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0002 |
| −3.3 | 0.0005 | 0.0005 | 0.0005 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0003 |
| −3.2 | 0.0007 | 0.0007 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0005 | 0.0005 | 0.0005 |
| −3.1 | 0.0010 | 0.0009 | 0.0009 | 0.0009 | 0.0008 | 0.0008 | 0.0008 | 0.0008 | 0.0007 | 0.0007 |
| −3.0 | 0.0013 | 0.0013 | 0.0013 | 0.0012 | 0.0012 | 0.0011 | 0.0011 | 0.0011 | 0.0010 | 0.0010 |
| −2.9 | 0.0019 | 0.0018 | 0.0018 | 0.0017 | 0.0016 | 0.0016 | 0.0015 | 0.0015 | 0.0014 | 0.0014 |
| −2.8 | 0.0026 | 0.0025 | 0.0024 | 0.0023 | 0.0023 | 0.0022 | 0.0021 | 0.0021 | 0.0020 | 0.0019 |
| −2.7 | 0.0035 | 0.0034 | 0.0033 | 0.0032 | 0.0031 | 0.0030 | 0.0029 | 0.0028 | 0.0027 | 0.0026 |
| −2.6 | 0.0047 | 0.0045 | 0.0044 | 0.0043 | 0.0041 | 0.0040 | 0.0039 | 0.0038 | 0.0037 | 0.0036 |
| −2.5 | 0.0062 | 0.0060 | 0.0059 | 0.0057 | 0.0055 | 0.0054 | 0.0052 | 0.0051 | 0.0049 | 0.0048 |
| −2.4 | 0.0082 | 0.0080 | 0.0078 | 0.0075 | 0.0073 | 0.0071 | 0.0069 | 0.0068 | 0.0066 | 0.0064 |
| −2.3 | 0.0107 | 0.0104 | 0.0102 | 0.0099 | 0.0096 | 0.0094 | 0.0091 | 0.0089 | 0.0087 | 0.0084 |
| −2.2 | 0.0139 | 0.0136 | 0.0132 | 0.0129 | 0.0125 | 0.0122 | 0.0119 | 0.0116 | 0.0113 | 0.0110 |
| −2.1 | 0.0179 | 0.0174 | 0.0170 | 0.0166 | 0.0162 | 0.0158 | 0.0154 | 0.0150 | 0.0146 | 0.0143 |
| −2.0 | 0.0228 | 0.0222 | 0.0217 | 0.0212 | 0.0207 | 0.0202 | 0.0197 | 0.0192 | 0.0188 | 0.0183 |
| −1.9 | 0.0287 | 0.0281 | 0.0274 | 0.0268 | 0.0262 | 0.0256 | 0.0250 | 0.0244 | 0.0239 | 0.0233 |
| −1.8 | 0.0359 | 0.0351 | 0.0344 | 0.0336 | 0.0329 | 0.0322 | 0.0314 | 0.0307 | 0.0301 | 0.0294 |
| −1.7 | 0.0446 | 0.0436 | 0.0427 | 0.0418 | 0.0409 | 0.0401 | 0.0392 | 0.0384 | 0.0375 | 0.0367 |
| −1.6 | 0.0548 | 0.0537 | 0.0526 | 0.0516 | 0.0505 | 0.0495 | 0.0485 | 0.0475 | 0.0465 | 0.0455 |
| −1.5 | 0.0668 | 0.0655 | 0.0643 | 0.0630 | 0.0618 | 0.0606 | 0.0594 | 0.0582 | 0.0571 | 0.0559 |
| −1.4 | 0.0808 | 0.0793 | 0.0778 | 0.0764 | 0.0749 | 0.0735 | 0.0721 | 0.0708 | 0.0694 | 0.0681 |
| −1.3 | 0.0968 | 0.0951 | 0.0934 | 0.0918 | 0.0901 | 0.0885 | 0.0869 | 0.0853 | 0.0838 | 0.0823 |
| −1.2 | 0.1151 | 0.1131 | 0.1112 | 0.1093 | 0.1075 | 0.1056 | 0.1038 | 0.1020 | 0.1003 | 0.0985 |
| −1.1 | 0.1357 | 0.1335 | 0.1314 | 0.1292 | 0.1271 | 0.1251 | 0.1230 | 0.1210 | 0.1190 | 0.1170 |
| −1.0 | 0.1587 | 0.1562 | 0.1539 | 0.1515 | 0.1492 | 0.1469 | 0.1446 | 0.1423 | 0.1401 | 0.1379 |
| −0.9 | 0.1841 | 0.1814 | 0.1788 | 0.1762 | 0.1736 | 0.1711 | 0.1685 | 0.1660 | 0.1635 | 0.1611 |
| −0.8 | 0.2119 | 0.2090 | 0.2061 | 0.2033 | 0.2005 | 0.1977 | 0.1949 | 0.1922 | 0.1894 | 0.1867 |
| −0.7 | 0.2420 | 0.2389 | 0.2358 | 0.2327 | 0.2296 | 0.2266 | 0.2236 | 0.2206 | 0.2177 | 0.2148 |
| −0.6 | 0.2743 | 0.2709 | 0.2676 | 0.2643 | 0.2611 | 0.2578 | 0.2546 | 0.2514 | 0.2483 | 0.2451 |
| −0.5 | 0.3085 | 0.3050 | 0.3015 | 0.2981 | 0.2946 | 0.2912 | 0.2877 | 0.2843 | 0.2810 | 0.2776 |
| −0.4 | 0.3446 | 0.3409 | 0.3372 | 0.3336 | 0.3300 | 0.3264 | 0.3228 | 0.3192 | 0.3156 | 0.3121 |
| −0.3 | 0.3821 | 0.3783 | 0.3745 | 0.3707 | 0.3669 | 0.3632 | 0.3594 | 0.3557 | 0.3520 | 0.3483 |
| −0.2 | 0.4207 | 0.4168 | 0.4129 | 0.4090 | 0.4052 | 0.4013 | 0.3974 | 0.3936 | 0.3897 | 0.3859 |
| −0.1 | 0.4602 | 0.4562 | 0.4522 | 0.4483 | 0.4443 | 0.4404 | 0.4364 | 0.4325 | 0.4286 | 0.4247 |
| −0.0 | 0.5000 | 0.4960 | 0.4920 | 0.4880 | 0.4840 | 0.4801 | 0.4761 | 0.4721 | 0.4681 | 0.4641 |

| $z$ | Area |
|---|---|
| −3.50 | 0.00023263 |
| −4.00 | 0.00003167 |
| −4.50 | 0.00000340 |
| −5.00 | 0.00000029 |

Source: Computed by M. Longnecker using Splus.

# Table A.1 Standard Normal Distribution Table




(Area under $z$ curve to the left of $z$)
$= P(Z < z) = P(Z \leq z)$
$= \Phi\ (z)$

| $z$ | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| 1.9 | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| 2.0 | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |
| 2.1 | 0.9821 | 0.9826 | 0.9830 | 0.9834 | 0.9838 | 0.9842 | 0.9846 | 0.9850 | 0.9854 | 0.9857 |
| 2.2 | 0.9861 | 0.9864 | 0.9868 | 0.9871 | 0.9875 | 0.9878 | 0.9881 | 0.9884 | 0.9887 | 0.9890 |
| 2.3 | 0.9893 | 0.9896 | 0.9898 | 0.9901 | 0.9904 | 0.9906 | 0.9909 | 0.9911 | 0.9913 | 0.9916 |
| 2.4 | 0.9918 | 0.9920 | 0.9922 | 0.9925 | 0.9927 | 0.9929 | 0.9931 | 0.9932 | 0.9934 | 0.9936 |
| 2.5 | 0.9938 | 0.9940 | 0.9941 | 0.9943 | 0.9945 | 0.9946 | 0.9948 | 0.9949 | 0.9951 | 0.9952 |
| 2.6 | 0.9953 | 0.9955 | 0.9956 | 0.9957 | 0.9959 | 0.9960 | 0.9961 | 0.9962 | 0.9963 | 0.9964 |
| 2.7 | 0.9965 | 0.9966 | 0.9967 | 0.9968 | 0.9969 | 0.9970 | 0.9971 | 0.9972 | 0.9973 | 0.9974 |
| 2.8 | 0.9974 | 0.9975 | 0.9976 | 0.9977 | 0.9977 | 0.9978 | 0.9979 | 0.9979 | 0.9980 | 0.9981 |
| 2.9 | 0.9981 | 0.9982 | 0.9982 | 0.9983 | 0.9984 | 0.9984 | 0.9985 | 0.9985 | 0.9986 | 0.9986 |
| 3.0 | 0.9987 | 0.9987 | 0.9987 | 0.9988 | 0.9988 | 0.9989 | 0.9989 | 0.9989 | 0.9990 | 0.9990 |
| 3.1 | 0.9990 | 0.9991 | 0.9991 | 0.9991 | 0.9992 | 0.9992 | 0.9992 | 0.9992 | 0.9993 | 0.9993 |
| 3.2 | 0.9993 | 0.9993 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9995 | 0.9995 | 0.9995 |
| 3.3 | 0.9995 | 0.9995 | 0.9995 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9997 |
| 3.4 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9998 |

| $z$ | Area |
|---|---|
| 3.50 | 0.99976737 |
| 4.00 | 0.99996833 |
| 4.50 | 0.99999660 |
| 5.00 | 0.99999971 |